

Path-Sensitive Small Divisors in Hahn–Fuchsian Systems

Exact convergence criteria, resonance amplification,
and stable analytic renormalization

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Abstract

A recent ProveIt manuscript characterizes universal absolute convergence of Hahn–Fuchsian normalizers when every exponent in a prescribed semigroup is an allowed input. It leaves open the corresponding problem for a smaller input support and restricted matrix directions. We resolve that problem for strictly upper-triangular perturbations of a real diagonal matrix, with independently variable coefficients on each permitted edge. The exact criterion involves only exponent sums along actual directed matrix paths: no endpoint spectral difference may be a left accumulation point of its path sumset. Accumulations created by algebraically impossible products are irrelevant. Exact resonances are allowed.

The proof gives an explicit polynomial-in-logarithm path kernel, an exact weighted multilinear norm criterion, and a conversion to the usual normalized log-free gauge. It also yields arbitrarily small, bounded-support counterexamples whenever the criterion fails. Two three-dimensional examples distinguish the path criterion from both a whole-semigroup test and an edge-by-edge test. An arbitrary-rank resonant chain exhibits a sharp loss of moments: coefficients c_n require exactly $\sum n^{r-1}|c_n| < \infty$ in dimension r , although the original input only requires $\sum |c_n| < \infty$.

For that chain, a stable basepoint-normalized analytic solution exists under the weaker input condition and realizes every fixed finite formal prefix. We prove a uniform $N \asymp \log(1/x)$ crossover with an explicit error bound. All specialized arguments are supplied. Exact finite checks and numerical illustrations accompany the article; they are not a Lean formalization. The research additions are relative to the inspected repository questions; worldwide priority has not been established.

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Notation at a glance

Symbol	Meaning
D, L	Euler derivation $x d/dx$ and the formal logarithm, $DL = 1$.
$\Lambda, \mathcal{G}$	Real diagonal constant matrix and the acyclic edge pattern.
E_p, ρ_p	Exponent sumset and spectral difference for a path p .
T_β, Q_p	Normalized polynomial inverse and its composed path kernel.
H, P	Canonical polynomial-log factor and normalized log-free gauge.
$M_p(w), \eta_p$	Weighted path norm and the nonresonant path gap.

1 The research question and the scope of the answer

1.1 A specific question in the repository

The ProveIt article *Finite Resonance Certificates and Sharp Analytic Normalization for Hahn–Fuchsian Systems* studies

$$xY' = (A_0 + A_+(x))Y$$

with positive, well-ordered real exponents. Its universal convergence theorem quantifies over all coefficient series supported in an entire exponent semigroup. Section 11 then asks whether that criterion remains necessary when the input is restricted to a smaller generating support. It explicitly suggests a triangular three-dimensional system with two permitted matrix units and an accumulating two-letter sumset [2].

This is not a cosmetic change of quantifiers. A semigroup remembers which exponents can be added, but forgets whether the associated matrix products are nonzero. For instance, the repeated product of a matrix unit E_{12} is zero, whereas $E_{12}E_{23} = E_{13}$ is not. A spectral accumulation arising only from the first kind of word need not obstruct normalization. Conversely, a bad sum formed by the second kind can be invisible in both individual coefficient series.

We solve this support-sensitive problem for every finite acyclic matrix pattern, not only the suggested three-dimensional example. After a permutation of vertices, such a pattern is strictly upper triangular. The constant term is diagonal with real eigenvalues; coefficients on different edges are independently variable. These are genuine restrictions, and the conclusions below retain them explicitly.

1.2 What is proved, and what is not

The principal results are as follows.

Question or issue	Result in this article
Smaller input supports and allowed matrix directions	An exact path–spectrum equivalence, including exact resonances; Theorems 6.1 and 7.3.
Weighted convergence when spectral gaps vanish	An exact necessary and sufficient path-kernel criterion for independent weighted ℓ^1 input classes; Theorem 5.1.
How severe can resonance amplification be?	In dimension r , an explicit chain has the exact moment threshold n^{r-1} ; Theorem 9.1.
What replaces a divergent canonical expansion?	A basepoint-normalized realization for the chain, with finite-prefix asymptotics and a uniform crossover certificate; Theorems 10.1 and 11.1.

We do not settle the smaller-support problem for arbitrary nontriangular matrices, fixed nonzero Jordan blocks, or coupled coefficient directions. We do not construct a multiplicative summation operation on an unrestricted Hahn field. The analytic realization theorem concerns the stated chain, not every formally divergent transseries equation. The distinction between universal independent inputs and correlated inputs is demonstrated explicitly in Section 8.3.

Editorial note (ProveIt, 2026-09-29). Theorem 6.1 specializes to the predecessor’s `thm:universal` [2] when every edge carries the whole exponent semigroup and the constant term is diagonal; that theorem also covers nontriangular perturbations and a nilpotent part, which are excluded here. The linear phenomenon of Example 8.2 has a nonlinear counterpart, a separated forcing whose products accumulate, in the later package `Nonlinear_Hahn_Fuchsian_Algebraic_Convergence_Loci/` (its `thm:hidden`); the nonlinear universal gap criterion is proved there and, independently, in `Nonlinear_Hahn_Dulac_Finite_Resonance_Control/`.

1.3 Relation to established theory

The Frobenius method, resonant Fuchsian normal forms, and the integration of nilpotent normal forms are classical; see Ilyashenko–Yakovenko [6, 7]. Well-based series and their differential calculus belong to the established theory of transseries [5]. Neither that general architecture nor the finite-dimensional inverse of $\beta + \partial_L$ is claimed as a new theorem.

Gontsov–Goryuchkina give a convergence theorem for generalized power-series solutions of polynomial ordinary differential equations; their displayed class has $\operatorname{Re} s_k \rightarrow +\infty$ [8]. Our coefficient data can instead have infinitely many exponents below a finite cut and are not assumed polynomial in x . The small-divisor work of Gontsov–Goryuchkina–Lastra concerns generalized solutions of q -difference equations [9]. It is relevant context, not a source of the graph-specific equivalence proved here.

The proposed additions are the sharp fixed-pattern criterion, its exact weighted version, the all-radii counterexample consequence, and the resonant-chain moment and crossover package. The earlier repository manuscript already supplies a whole-semigroup criterion, finite resonance ancestry, and a rank-one accumulation model. We build on its questions rather than relabel those results as new.

1.4 Provenance and reading boundary

Repository comparisons use snapshot 9250bbf8af80dacf7b252d9d0320b122c80961ba. We inspected the series group index, the Hahn–Fuchsian package README, the article’s setup and normal-form section and its further-questions section, the resonance-block package README, and the well-based-support Lean module [1, 2, 4, 3]. The complete large canonical volume and every other research arrival were not audited. This is a targeted comparison, not a claim of exhaustive novelty checking across the repository or literature. No repository files were changed.

2 Fixed supports, directed paths, and coefficient spaces

2.1 The differential system

Fix $r \geq 2$, real numbers $\lambda_1, \dots, \lambda_r$, and

$$\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_r), \quad D = x \frac{d}{dx}.$$

Let $\mathcal{G}$ be a directed graph on $\{1, \dots, r\}$ whose edges have the form $i \rightarrow j$ with $i < j$. For every edge $e = (i, j)$ fix a nonempty well-ordered set $E_e \subset \mathbb{R}_{>0}$. A missing or empty edge is simply deleted. Consider

$$DY = (\Lambda + A(x))Y, \quad A_{ij}(x) = \sum_{\alpha \in E_{ij}} a_{ij,\alpha} x^\alpha, \quad (2.1)$$

with $A_{ij} = 0$ off the prescribed edges. Coefficients are complex and may vanish. The support sets, graph, and spectrum are fixed; the coefficients are the variable data.

We work formally in

$$\mathcal{K} = \mathbb{C}((x^{\mathbb{R}})), \quad \mathcal{K}[L], \quad D(x^\alpha) = \alpha x^\alpha, \quad DL = 1.$$

Here a Hahn support is well ordered in the ordinary order of real exponents. The variable L is algebraically independent formally and is later evaluated as $\log x$. Smaller powers are dominant as $x \rightarrow 0$.

For a path $p = (i_0, i_1, \dots, i_m)$ put

$$E_p = E_{i_0 i_1} + \dots + E_{i_{m-1} i_m}, \quad \rho_p = \lambda_{i_0} - \lambda_{i_m}. \quad (2.2)$$

The length m is at most $r - 1$. Write S_{ij} for the finite union of all E_p with endpoints i, j . No full additive semigroup is needed to describe these supports.

Lemma 2.1 (Finite path support arithmetic). *Every E_p and S_{ij} is well ordered. For a fixed path and a fixed real s , there are only finitely many tuples $(\alpha_1, \dots, \alpha_m)$, with each α_k in its edge support, whose sum is s .*

Proof. For two well-ordered subsets U, V of a linearly ordered abelian group, $U + V$ is well ordered and every fixed sum has finitely many decompositions. In the present real case this can also be seen directly. Every sequence in a well-ordered set has a nondecreasing subsequence. A strictly decreasing sequence $u_n + v_n$ would therefore, after taking such a subsequence of the u_n , give a strictly decreasing sequence of v_n , a contradiction. If infinitely many distinct pairs had the same sum, their distinct first coordinates would have a strictly increasing subsequence, forcing strictly decreasing second coordinates. Induction proves both assertions for any fixed number of factors. There are only finitely many paths in the graph, and a finite union of well-ordered subsets of a linear order is well ordered. $\square$

The elementary subsequence fact used here follows by choosing successive tail minima; equivalently, a sequence with no nondecreasing subsequence would yield an infinite strictly decreasing subsequence. The repository provides the corresponding two-factor support and finite-antidiagonal bridges as `Fabius.neumann_add_isPWO` and `Fabius.neumann_finite_decompositions` [3].

2.2 Divided powers and absolute coefficient norms

Put $L^{[k]} = L^k/k!$. For $u(L) = \sum_{k=0}^d u_k L^{[k]}$ define

$$|u|_{\text{dp}} = \sum_{k=0}^d |u_k|. \quad (2.3)$$

On a fixed degree bound, differentiation shifts the divided-power coefficients and has norm at most one. Integration with zero constant coefficient also has norm at most one, as a map to the next degree.

For $R > 0$, a fixed well-ordered E , and a positive finite weight $w : E \rightarrow (0, \infty)$, let

$$X_R(E, w) = \left\{ a : \sum_{\alpha \in E} |a_\alpha| R^\alpha w(\alpha) < \infty \right\}. \quad (2.4)$$

This is a Banach space, isometric to ordinary $\ell^1(E)$ after multiplying coordinates by $R^\alpha w(\alpha)$. For a fixed output support S and degree bound d , use

$$\|h\|_{R, \text{dp}} = \sum_{s \in S} R^s |h_s(L)|_{\text{dp}}, \quad h = \sum_{s \in S} h_s(L) x^s. \quad (2.5)$$

The resulting space is likewise Banach. Subsets of these fixed supports remain Hahn supports, so completion does not leave the formal class.

Let $\mathcal{A}_R$ denote the collection of Hahn series with finite norm $\sum |f_s| R^s$, allowing any individual well-ordered support. It is a normed algebra under finite coefficient convolution. We do *not* claim that this unrestricted-support algebra is complete: an ℓ^1 limit of changing supports need not have well-ordered support. All closed-graph arguments below take place in the fixed-support Banach spaces just defined.

Membership in $\mathcal{A}_R[L]$ means that each of finitely many log coefficients belongs to $\mathcal{A}_R$. Multiplication by a finite sum of real monomials times log polynomials preserves this membership. Indeed, multiplication by x^β scales the coefficient norm by R^β , and multiplication of fixed-degree polynomials changes norms by only a finite combinatorial constant. No completeness assertion is required for this observation.

3 The exact path kernel

3.1 A polynomial inverse including exact resonance

For $\beta \in \mathbb{R}$ define a linear operator on $\mathbb{C}[L]$ by

$$T_\beta u = \begin{cases} \sum_{k=0}^{\deg u} \frac{(-1)^k}{\beta^{k+1}} \partial_L^k u, & \beta \neq 0, \\ \int_0^L u(v) dv, & \beta = 0. \end{cases} \quad (3.1)$$

The zero polynomial is mapped to zero.

Lemma 3.1 (Polynomial right inverse). *For every u , $(\beta + \partial_L)T_\beta u = u$. If $\beta \neq 0$, $T_\beta u$ is the unique polynomial solution. If $\beta = 0$, it is the unique polynomial solution with zero constant coefficient. For $u \neq 0$, a nonzero β preserves degree and divides the divided-power leading coefficient by β ; T_0 raises degree by one and preserves that leading coefficient.*

Proof. For nonzero β the finite geometric sum telescopes, because a sufficiently high derivative of u is zero. A nonzero polynomial cannot lie in the kernel of $\beta + \partial_L$: its highest-degree coefficient would be multiplied by the nonzero scalar β . For $\beta = 0$, differentiation and zero-constant integration are inverse in the stated sense. The leading-coefficient assertions follow directly in the $L^{[k]}$ basis. $\square$

For degree at most d and $\beta \neq 0$,

$$|T_\beta u|_{\text{dp}} \leq \left(\sum_{k=0}^d |\beta|^{-k-1} \right) |u|_{\text{dp}}. \quad (3.2)$$

In particular, a uniform lower bound on nonzero $|\beta|$ bounds all operators needed for paths of bounded length.

3.2 Suffix spectral differences

Fix a path $p = (i_0, \dots, i_m)$ and an exponent tuple $\alpha = (\alpha_1, \dots, \alpha_m)$. Define

$$\beta_k = \alpha_k + \dots + \alpha_m - (\lambda_{i_{k-1}} - \lambda_{i_m}), \quad 1 \leq k \leq m, \quad (3.3)$$

and the polynomial path kernel

$$Q_p(\alpha; L) = T_{\beta_1} T_{\beta_2} \cdots T_{\beta_m} 1. \quad (3.4)$$

The rightmost operator is applied first. The spectral differences refer to the *same terminal vertex*. Replacing them by the individual edge differences is generally incorrect.

Lemma 3.2 (Degree and nonvanishing of a path kernel). *Let z be the number of zero entries among $\beta_1, \dots, \beta_m$. Then*

$$\deg_L Q_p = z, \quad \text{lc}_{\text{dp}} Q_p = \prod_{\beta_k \neq 0} \beta_k^{-1}. \quad (3.5)$$

In particular, no path kernel is the zero polynomial.

Proof. Start with the polynomial 1 and apply Lemma 3.1 from right to left. Every exact resonance raises degree by one without changing the divided-power leading coefficient. Every other step multiplies that coefficient by β_k^{-1} . These factors are nonzero. $\square$

This formula is useful for the necessity direction. It avoids attempting to estimate a potentially cancellation-prone constant coefficient of a log polynomial. The highest divided-power coefficient is a single product.

4 A canonical Frobenius fundamental matrix

We first construct a polynomial-logarithmic factor H such that $Y = Hx^\Lambda$ solves (2.1). The log-free normalizer is denoted by P and constructed separately in Section 7; keeping the two letters distinct prevents a change of normalization from being hidden.

Theorem 4.1 (Canonical path expansion). *There is a unique upper-unitriangular matrix $H \in \text{Mat}_r(\mathcal{K}[L])$ with positive off-diagonal support that satisfies*

$$DH = [\Lambda, H] + AH \quad (4.1)$$

and the following normalization: whenever $s = \lambda_i - \lambda_j > 0$, the polynomial coefficient $[x^s]H_{ij}$ has zero constant coefficient. Its entries are

$$H_{ij} = \sum_{p \in \text{Paths}(i,j)} \sum_{\alpha_k \in E_{i_{k-1}i_k}} \left(\prod_{k=1}^m a_{i_{k-1}i_k, \alpha_k} \right) x^{\alpha_1 + \dots + \alpha_m} Q_p(\alpha; L) \quad (i < j). \quad (4.2)$$

Every coefficient sum is finite, $\text{supp } H_{ij} \subseteq S_{ij}$, and $\deg_L H_{ij} \leq j - i \leq r - 1$. Thus Hx^Λ is a formal fundamental matrix.

Proof. Proceed by increasing $j - i$. With $H_{jj} = 1$, the (i, j) equation is

$$(D - \rho_{ij})H_{ij} = A_{ij} + \sum_{i < k < j} A_{ik}H_{kj}, \quad \rho_{ij} = \lambda_i - \lambda_j. \quad (4.3)$$

Its right side has already been defined. On a coefficient $x^s u(L)$, the left side acts as $x^s(s - \rho_{ij} + \partial_L)u$. Apply $T_{s - \rho_{ij}}$ to each coefficient. Lemma 3.1 gives exactly the stated normalization and uniqueness. Positivity of input exponents and Lemma 2.1 justify the coefficient operations.

Expanding the recursion prepends one edge to a suffix path. The homological factor at that step is the corresponding β_k in (3.3). Induction therefore gives (4.2). There are at most $j - i$ edges in a path, and each T_0 raises degree by at most one. The determinant of H is one, so Hx^Λ is invertible. $\square$

The theorem requires neither analytic convergence nor local finiteness of the supports. Infinite sets below a finite spectral value do not invalidate the formal recursion: a fixed coefficient still has only finitely many contributing tuples.

5 An exact weighted convergence criterion

5.1 The path norm

Assign a weight $w_e : E_e \rightarrow (0, \infty)$ to each edge. For a path p define

$$M_p(w) = \sup_{\alpha \in \prod E_e} \frac{|Q_p(\alpha; L)|_{\text{dp}}}{\prod_{e \in p} w_e(\alpha_e)} \in [0, \infty]. \quad (5.1)$$

The exponent-radius factors do not occur: $R^{\sum \alpha_e}$ cancels against $\prod R^{\alpha_e}$.

Theorem 5.1 (Exact independent-input criterion). *Fix the graph, supports, spectrum, weights, and $R > 0$. The following are equivalent:*

- (i) *Every independent choice $a_e \in X_R(E_e, w_e)$ gives a canonical H with all coefficients in $\mathcal{A}_R[L]$.*

(ii) $M_p(w) < \infty$ for every directed path p .

If these conditions hold, then

$$\|H_{ij}\|_{R,dp} \leq \sum_{p \in \text{Paths}(i,j)} M_p(w) \prod_{e \in p} \|a_e\|_{X_R(E_e, w_e)}. \quad (5.2)$$

Moreover, $M_p(w)$ is exactly the multilinear operator norm of the isolated path contribution, not merely an upper bound.

Sufficiency and exact norm. For a fixed path, the triangle inequality, (5.1), and the identity $R^{\sum \alpha_e} = \prod R^{\alpha_e}$ bound the output norm by $M_p(w)$ times the product of the input norms. Absolute summability justifies the resulting product of scalar sums. Adding the finitely many paths proves (5.2).

To see sharpness, give each edge a single nonzero coefficient at α_e , with magnitude $(R^{\alpha_e} w_e(\alpha_e))^{-1}$. Every input has norm one and the output norm is exactly the ratio in (5.1). Taking the supremum proves the reverse norm inequality. $\square$

The necessity direction needs more than observing that single-coefficient examples have arbitrarily large norms. A map can be unbounded on a sequence of finite inputs without that observation alone producing one infinite input with divergent output. The following standard functional-analytic argument supplies the missing quantifier.

Lemma 5.2 (Automatic boundedness for the path map). *Suppose the formal multilinear contribution of a fixed path maps every tuple of its fixed-support Banach input spaces into the fixed-support Banach output space of degree at most $r - 1$. Then that multilinear map is bounded.*

Proof. Fix all but one input. Consider a sequence in the remaining input space converging in norm, whose outputs also converge in output norm. Every coefficient of a fixed exponent and fixed divided power is a finite sum of input coordinates by Lemma 2.1. Coordinate evaluation is continuous in each weighted ℓ^1 space. Thus the output limit has the same coordinates as the formal output of the input limit. The graph of the resulting everywhere-defined linear map is closed. The closed graph theorem makes it continuous.

The multilinear map is therefore separately continuous. Separate continuity on a finite product of Banach spaces implies joint boundedness. For completeness, argue by induction on the number of factors. Fixing the first factor, the induction hypothesis bounds the map on the unit balls of the remaining factors. Regard those remaining arguments as indexing a family of bounded linear maps of the first factor. That family is pointwise bounded by the induction hypothesis. The uniform boundedness principle supplies a uniform operator bound, as required. $\square$

Necessity in Theorem 5.1. Fix a path and set every edge not on that path equal to zero. Between its endpoints, the retained graph has just that full path. Assumption (i) therefore says that its formal multilinear contribution maps every tuple of edge inputs into the fixed-support output Banach space. By Lemma 5.2 it has finite norm. The single-coefficient tests in the first part of the proof show that this norm is $M_p(w)$. $\square$

Remark 5.3 (What the word “exact” means). The criterion is exact for the specified independent product of weighted ℓ^1 spaces and for absolute coefficient convergence. It is not a necessary test for one selected input with cancellations, nor for a family whose edge coefficients satisfy additional algebraic relations. Nor does it claim that the infinite supremum is computable from every possible presentation of the supports and weights.

6 The path–spectrum gap theorem

For every path define its nonresonant gap

$$\eta_p = \inf\{|s - \rho_p| : s \in E_p, s \neq \rho_p\}, \quad (6.1)$$

with infimum $+\infty$ for an empty set. Exact resonance is explicitly removed from the infimum.

Theorem 6.1 (Sharp accessible-spectrum criterion). *For the system (2.1), the following conditions are equivalent:*

- (i) *For one fixed $R > 0$, every input with $\sum_{e,\alpha} |a_{e,\alpha}| R^\alpha < \infty$ has $H \in \text{Mat}_r(\mathcal{A}_R[L])$.*
- (ii) *The same assertion holds for every $R > 0$.*
- (iii) *$\eta_p > 0$ for every directed path p .*
- (iv) *No ρ_p is a left accumulation point of its own path sumset E_p .*

Thus a spectral difference matters only for paths with its specified endpoints. It is not enough to take an undifferentiated union of all path sumsets and all spectral differences.

Proof. Assume (iii). There are finitely many paths, so every nonzero β_k in every path tuple is bounded away from zero by one common $\eta > 0$. Indeed, a suffix is itself a path, and its β_k is exactly a nonresonant endpoint difference for that suffix. Let $d = r - 1$ and $h = \min(1, \eta)$. Equations (3.2) and $|T_0 u|_{\text{dp}} = |u|_{\text{dp}}$ give the uniform bound

$$|Q_p|_{\text{dp}} \leq ((d+1)h^{-(d+1)})^{|p|}.$$

Consequently every unweighted M_p is finite. Theorem 5.1 proves (ii), which immediately implies (i).

Conversely, suppose $\eta_p = 0$ for a path of length m . Choose tuples with sums $s_n \neq \rho_p$ and $s_n \rightarrow \rho_p$. Their first homological factors satisfy $\beta_{1,n} = s_n - \rho_p \rightarrow 0$. Because all input exponents are positive and s_n is bounded, every suffix sum is bounded. Hence there is a fixed $B \geq 1$ such that every $|\beta_{k,n}| \leq B$. By Lemma 3.2, the highest divided-power coefficient of the kernel satisfies

$$|Q_p(\alpha_n)|_{\text{dp}} \geq \frac{1}{B^{m-1}|s_n - \rho_p|} \rightarrow \infty. \quad (6.2)$$

Zero suffix factors cause no problem: they are omitted from the leading coefficient product. Thus $M_p = \infty$, contradicting (i) by Theorem 5.1. This proves (i) $\Rightarrow$ (iii).

Finally, a well-ordered real set cannot accumulate at a finite point from above: one could successively choose smaller elements approaching that point and obtain an infinite decreasing sequence. If its distance from a point, with exact equality removed, is zero, it therefore accumulates from below. The reverse implication is immediate. Apply this to each E_p to obtain (iii) $\Leftrightarrow$ (iv). $\square$

Corollary 6.2 (Small inputs; no rescue by reducing the radius). *If the gap condition fails, then for every $\varepsilon > 0$ there is an input supported on a single directed chain, with all input exponents in a bounded interval and $\sum_{e,\alpha} |a_{e,\alpha}| < \varepsilon$, whose canonical H fails absolute coefficient convergence at every radius $R > 0$.*

Proof. Use a bad path from the proof of Theorem 6.1. All exponents in its witnessing tuples lie below one finite B_0 . Restrict each edge to $E_e \cap (0, B_0]$. The corresponding path norm remains infinite, so the necessity argument in Theorem 5.1, now at radius one, provides a tuple of ℓ^1 inputs whose path output is not in ℓ^1 . Set all other edges to zero. Every output exponent lies in $(0, (r-1)B_0]$. On a bounded interval, the weights R^s are bounded above and below by

positive constants, so nonconvergence persists at every radius. Scaling all retained inputs by one sufficiently small positive number scales this isolated path contribution by a nonzero power of that number and makes the total input norm less than ε . $\square$

For H , the conclusion refers to its polynomial-logarithmic coefficient norm, not to every possible exceptional numerical value of L . The next section transfers it to a genuinely log-free gauge, where absolute convergence at radius R is the ordinary absolute generalized-series test.

7 Equivalence with the normalized log-free gauge

7.1 Triangular normal form without an analytic assumption

For $i < j$ put $\rho_{ij} = \lambda_i - \lambda_j$. We seek an upper-unitriangular log-free matrix P and a strictly upper-triangular resonant matrix $\mathcal{R}$ satisfying

$$DP = (\Lambda + A)P - P(\Lambda + \mathcal{R}), \quad (7.1)$$

where

$$\mathcal{R}_{ij} = b_{ij}x^{\rho_{ij}} \quad \text{if } \rho_{ij} > 0, \quad \mathcal{R}_{ij} = 0 \quad \text{otherwise,} \quad (7.2)$$

and $[x^{\rho_{ij}}]P_{ij} = 0$ whenever $\rho_{ij} > 0$.

Proposition 7.1 (Unique normalized log-free pair). *There is a unique pair $(P, \mathcal{R})$ of this form with positive off-diagonal support. The support of each entry is contained in its path sumset union. The matrix $B = (b_{ij})$ is constant and nilpotent, and*

$$F = Px^\Lambda e^{BL} \quad (7.3)$$

is a formal fundamental matrix. Every coefficient of B depends polynomially on finitely many original input coefficients, with constants determined by the fixed support and spectral data.

Proof. Proceed again by increasing $j - i$. Define, using previously computed entries,

$$G_{ij} = A_{ij} + \sum_{i < k < j} A_{ik}P_{kj} - \sum_{i < k < j} P_{ik}\mathcal{R}_{kj}. \quad (7.4)$$

If $\rho_{ij} > 0$, put $b_{ij} = [x^{\rho_{ij}}]G_{ij}$; otherwise put $b_{ij} = 0$. Set

$$P_{ij} = \sum_{s \neq \rho_{ij}} \frac{[x^s]G_{ij}}{s - \rho_{ij}} x^s. \quad (7.5)$$

All supports are positive, so there is no omitted nonpositive resonance. This is exactly the coefficient equation for $(D - \rho_{ij})P_{ij} = G_{ij} - \mathcal{R}_{ij}$ and proves existence and uniqueness.

Every product in (7.4) concatenates two permitted paths. Consequently its support remains within S_{ij} . Lemma 2.1 then proves finite coefficient dependence at each fixed resonance. Recursion uses only additions, multiplications, and division by fixed nonzero real numbers, so the dependence on input coefficients is polynomial. There are finitely many matrix entries.

Since B is strictly upper triangular, $B^r = 0$. Equation (7.2) is $\mathcal{R} = x^\Lambda B x^{-\Lambda}$. Differentiating (7.3) and using (7.1) now proves the fundamental-matrix assertion. $\square$

The finite-dependence statement is compatible with, and not a replacement for, the more general finite-ancestry result in the predecessor [2]. Our triangular construction is included to make the normalization comparison self-contained.

7.2 Why the two convergence questions are the same

Lemma 7.2 (Constants). *The constants of $\mathcal{K}[L]$ under D are exactly $\mathbb{C}$.*

Proof. A constant Hahn series has only exponent zero, since its coefficient at x^s is multiplied by s under D . Also, no derivative of a Hahn series has a nonzero x^0 coefficient. If a nonzero polynomial $\sum_{k=0}^d f_k L^k$ had derivative zero and $d > 0$, its highest coefficient would satisfy $Df_d = 0$ and hence be a nonzero complex constant. The next equation would be $Df_{d-1} = -df_d$, contradicting the missing x^0 coefficient. Thus $d = 0$. $\square$

Theorem 7.3 (Convergence is independent of these two normalizations). *For every fixed formal input and every $R > 0$,*

$$H \in \text{Mat}_r(\mathcal{A}_R[L]) \iff P \in \text{Mat}_r(\mathcal{A}_R). \quad (7.6)$$

Consequently Theorem 6.1 is an exact universal convergence criterion for the normalized log-free gauge P . The weighted membership criterion of Theorem 5.1 also transfers to P for the same weighted input classes. If the gap condition fails, the small input in Corollary 6.2 has a log-free normalizer diverging absolutely at every positive radius.

Proof. Both $F = Px^\Lambda e^{BL}$ and $Y = Hx^\Lambda$ are invertible fundamental matrices over $\mathcal{K}[L]$. Their inverses lie in the same ring: upper unitriangular inverses and e^{-BL} are finite sums, and the diagonal monomials are invertible. Therefore $C = F^{-1}Y$ has derivative zero and belongs to $\text{GL}_r(\mathbb{C})$ by Lemma 7.2. We obtain

$$H = Px^\Lambda e^{BL} C x^{-\Lambda}, \quad P = H x^\Lambda C^{-1} e^{-BL} x^{-\Lambda}. \quad (7.7)$$

Each rightmost factor is a finite sum of monomials times log polynomials. Multiplication by it preserves $\mathcal{A}_R[L]$ membership. Since P is log-free, (7.6) follows. The universal and all-radii conclusions are then immediate. $\square$

This is an equivalence of membership for each input. It does not identify the path norm of H with an identical operator norm of P , because the finite factors B and C depend on the input. The exact norm assertion in Theorem 5.1 remains an assertion about the isolated polynomial-logarithmic path operator.

On every smaller radius $0 < R' < R$, an absolutely convergent gauge evaluates to an analytic function on a chosen logarithmic cover. Differentiation is legitimate there: multiplying coefficients by s^k is controlled by the exponential margin $(R'/R)^s$, while all supports under consideration are bounded below. The formal equations therefore become analytic equations. No differentiability assertion at the outer boundary $|x| = R$ is needed.

8 What three and four dimensions already show

8.1 A bad semigroup accumulation that cannot occur in a matrix word

Example 8.1 (The whole-semigroup condition is not necessary). Take

$$\Lambda = \text{diag}(3, 1, 0), \quad E_{12} = \{3/2 - 1/n : n \geq 2\}, \quad E_{23} = \{10\},$$

with no other edges. The three permitted paths have data

p	E_p	ρ_p
$1 \rightarrow 2$	E_{12}	2
$2 \rightarrow 3$	$\{10\}$	1
$1 \rightarrow 2 \rightarrow 3$	$E_{12} + 10$	3.

Their gaps are at least $1/2$, 9 , and 8 , respectively. Thus every absolutely convergent input has an absolutely convergent normalized gauge. Yet the full semigroup generated by $E_{12} \cup \{10\}$ contains $3 - 2/n \rightarrow 3$ from below. The positive spectral difference 3 is therefore a left accumulation point of that semigroup.

Here $2E_{12}$ describes a forbidden word, because $E_{12}E_{12} = 0$ as a matrix product. Writing $A_{12} = \sum a_\alpha x^\alpha$ and $A_{23} = bx^{10}$, the actual nonresonant entries are

$$P_{12} = \sum \frac{a_\alpha}{\alpha - 2} x^\alpha, \quad P_{23} = \frac{b}{9} x^{10}, \quad P_{13} = \sum \frac{a_\alpha b}{9(\alpha + 7)} x^{\alpha+10}.$$

Every denominator is uniformly safe. This answers the necessity question negatively if it is phrased using the full generated semigroup without matrix-path information.

8.2 Safe edges whose composition is unsafe

Example 8.2 (An accumulating two-letter sumset). Take

$$\Lambda = \text{diag}(3, 2, 0), \quad A_{12} = x^{1/2}, \quad A_{23} = \sum_{n \geq 3} n^{-p} x^{5/2-1/n}, \quad p > 1.$$

The individual gaps are $1/2$ and at least $1/6$. Nevertheless, $1/2 + (5/2 - 1/n) = 3 - 1/n \rightarrow 3$, the endpoint difference for the composed path. There are no exact resonances, and

$$\begin{aligned} P_{12} &= -2x^{1/2}, \\ P_{23} &= \sum_{n \geq 3} \frac{n^{-p}}{1/2 - 1/n} x^{5/2-1/n}, \\ P_{13} &= - \sum_{n \geq 3} \frac{n^{1-p}}{1/2 - 1/n} x^{3-1/n}. \end{aligned} \tag{8.1}$$

Thus the original input and both edge normalizations converge for $p > 1$, whereas the full normalizer converges exactly when $p > 2$.

The equivalence at every radius follows because all displayed exponent sets are bounded. The coefficients of P_{13} have one sign, so there is no cancellation issue. This is the specific two-letter phenomenon proposed in the predecessor’s further-questions section, now placed inside a necessary-and-sufficient theorem for all acyclic patterns.

8.3 Why independent inputs cannot be silently replaced by coupled inputs

Take the same $f(x) = \sum_{n \geq 3} n^{-p} x^{5/2-1/n}$, and set

$$\Lambda = \text{diag}(3, 2, 2, 0), \quad A_{12} = A_{13} = x^{1/2}, \quad A_{24} = f, \quad A_{34} = -f,$$

with all other entries zero. The two paths from 1 to 4 have identical kernels and opposite coefficient products. Hence $H_{14} = P_{14} = 0$. The remaining nonzero entries have the safe one-edge denominators in Example 8.2, and the whole system normalizes convergently for $p > 1$ even though each of the two path sumsets accumulates at 3.

This does not contradict Theorem 6.1: the selected coefficients are correlated. In matrix-direction language, $(E_{12} + E_{13})(E_{24} - E_{34}) = 0$. A graph that records the four edges but forgets this relation loses essential information. The exact weighted criterion handles all independent inputs; a genuinely optimal criterion for constrained matrix directions must retain the corresponding algebraic relations among words.

9 Resonant chains and sharp arbitrary-order moment loss

9.1 The family

For $r \geq 2$, let

$$\lambda_i = r - i, \quad A_{12} = \sum_{n \geq 2} c_n x^{1-1/n}, \quad A_{j,j+1} = x \quad (2 \leq j < r), \quad (9.1)$$

with no other edges. The first edge approaches its resonance from below; every downstream edge is exactly resonant. Put $q = r - 2$.

Theorem 9.1 (Exact moment threshold). *For every $R > 0$, the input in (9.1) is absolutely convergent exactly when $\sum |c_n| < \infty$. Its normalized log-free gauge is*

$$P_{1j} = - \sum_{n \geq 2} c_n n^{j-1} x^{j-1-1/n} \quad (2 \leq j \leq r), \quad (9.2)$$

with all other off-diagonal entries zero. Its resonant constant matrix is $B_{j,j+1} = 1$ for $2 \leq j < r$, with all other entries zero. The full gauge converges absolutely if and only if

$$\sum_{n \geq 2} |c_n| n^{r-1} < \infty. \quad (9.3)$$

In particular, for $c_n = n^{-p}$ the input converges for $p > 1$, while the normalizer converges exactly for $p > r$.

Proof. For P_{12} , division by $(1 - 1/n) - 1 = -1/n$ gives (9.2). The downstream input edges are placed entirely in $\mathcal{R}$, and their gauge entries are zero. For $j > 2$, the only nonzero term in (7.4) is $-P_{1,j-1} \mathcal{R}_{j-1,j} = -x P_{1,j-1}$. Dividing again by $-1/n$ multiplies the previous coefficient by n . Induction proves the formula and the stated B .

For fixed j and $R > 0$, the factors $R^{j-1-1/n}$ are bounded above and below by positive constants. Thus convergence of P_{1j} is equivalent to $\sum |c_n| n^{j-1} < \infty$. The strongest condition occurs at $j = r$. The same bounded-exponent observation proves the input assertion. Finally, $\sum n^{r-1-p}$ converges exactly when $p > r$. $\square$

The loss can therefore be made arbitrarily large by increasing the nilpotent chain length. A single division by the first small denominator does not measure the full cost of normalization.

9.2 The logarithmic explanation

For $2 \leq j \leq r$,

$$H_{jr} = x^{r-j} \frac{L^{r-j}}{(r-j)!}.$$

For a fixed n in the first edge, the remaining q homological factors are zero, so the top-right kernel is

$$T_{-1/n} L^{[q]} = - \sum_{k=0}^q n^{k+1} L^{[q-k]}. \quad (9.4)$$

Consequently

$$H_{1r} = - \sum_{n \geq 2} c_n x^{q+1-1/n} \sum_{k=0}^q n^{k+1} \frac{L^{q-k}}{(q-k)!}. \quad (9.5)$$

Its divided-power coefficient norm is exactly $\sum |c_n| R^{q+1-1/n} \sum_{k=0}^q n^{k+1}$, confirming the same moment condition.

For independent weighted inputs on the first edge, with the downstream coefficients allowed to vary over their one-dimensional spaces, the minimal weight growth is therefore $w(1 - 1/n) \asymp n^{r-1}$: more precisely, universal convergence holds exactly when $\sup_{n \geq 2} n^{r-1}/w(1 - 1/n) < \infty$. This is an exact consequence of the path norm, not a guessed submultiplicative majorant.

10 Stable analytic realization beyond the moment threshold

10.1 A convergent kernel instead of divergent constants

Assume only $\sum_{n \geq 2} |c_n| < \infty$. Put $t = \log(1/x)$, initially for $0 < x \leq 1$. Define the entire function

$$\Psi_q(z) = \frac{1}{q!} \int_0^1 e^{zu} u^q du = \frac{1}{q!} \sum_{k \geq 0} \frac{z^k}{k!(q+k+1)}. \quad (10.1)$$

The integral or the positive series is a stable way to evaluate it near $z = 0$.

Theorem 10.1 (Basepoint-normalized realization). *The chain (9.1) has a unique analytic fundamental matrix on the logarithmic cover with $Y(1) = I$. Writing it as $Y^{\text{bp}} = H^{\text{bp}} x^\Lambda$, its top-right entry is*

$$H_{1r}^{\text{bp}}(e^{-t}) = \frac{(-1)^{q+1} e^{-(q+1)t}}{q!} \sum_{n \geq 2} c_n \int_0^t e^{v/n} v^q dv \quad (10.2)$$

$$= (-1)^{q+1} e^{-(q+1)t} t^{q+1} \sum_{n \geq 2} c_n \Psi_q(t/n). \quad (10.3)$$

The analogous formula with $q = j - 2$ gives H_{1j}^{bp} . All downstream entries are the finite resonant expressions already given. These functions are entire in t and satisfy the differential system.

Moreover, this analytic matrix realizes every fixed finite exponent-block prefix of the canonical expansion, even when (9.3) fails.

Proof. For downstream entries, direct integration gives $x^{k-j} L^{k-j}/(k-j)!$. For the top-right entry, variation of constants in (4.3), with value zero at $x = 1$, gives

$$H_{1r}^{\text{bp}}(x) = x^{q+1} \sum_{n \geq 2} c_n \int_0^{\log x} e^{-u/n} \frac{u^q}{q!} du.$$

The substitution $u = -v$ proves (10.2), and $v = tu$ gives (10.3). For complex t in a compact set, $|\Psi_q(t/n)| \leq e^{|t|/2}/(q+1)!$. Hence the sum converges normally under $\sum |c_n| < \infty$, and the same holds after any fixed number of t derivatives. The equations and the basepoint normalization follow termwise. Triangular variation of constants also proves uniqueness.

It remains to justify the formal-prefix assertion. For each fixed n , elementary integration gives, with $E_q(z) = \sum_{j=0}^q z^j/j!$,

$$H_{1r,n}^{\text{bp}}(e^{-t}) = c_n e^{-(q+1)t} n^{q+1} [1 - e^{t/n} E_q(-t/n)]. \quad (10.4)$$

The term containing $e^{t/n} E_q(-t/n)$ is exactly the n th canonical block in (9.5); the remaining term has exponent $q+1$. Fix $M \geq 2$ and subtract the canonical blocks $2 \leq n \leq M$. The finitely many added exponent- $(q+1)$ terms are $O(x^{q+1})$. For the remaining basepoint terms and $t \geq 1$,

$$\begin{aligned} \sum_{n > M} |H_{1r,n}^{\text{bp}}(e^{-t})| &\leq \frac{e^{-(q+1)t}}{q!} \sum_{n > M} |c_n| \int_0^t e^{v/(M+1)} v^q dv \\ &\leq \frac{M+1}{q!} \|c\|_{\ell^1} e^{-(q+1-1/(M+1))t} t^q. \end{aligned}$$

Thus the remainder is $O(x^{q+1-1/(M+1)}t^q)$. Its exponent is strictly larger than the last retained exponent $q + 1 - 1/M$, so it is smaller than that block's monomial times every fixed inverse power of t . The same proof applies to each top-row entry. Downstream expansions are finite. $\square$

Formula (10.4) must be kept grouped when the relevant moments diverge. Splitting it into two separately summed expressions would introduce the divergent quantity $\sum c_n n^{q+1}$. The stable expression (10.3) never forms that quantity. Indeed,

$$1 - e^z E_q(-z) = (-1)^{q+1} z^{q+1} \Psi_q(z),$$

so precisely $q + 1$ powers of the small parameter cancel.

This is a selected analytic realization determined by a basepoint. It is not an assertion that the original canonical Hahn sum converges, or that formal series without a chosen analytic normalization determine unique functions. When the strong moment condition holds, the canonical and basepoint fundamental matrices are related by an ordinary finite constant matrix. Below that condition, their formal comparison cannot be justified by summing the divergent constants separately.

11 A uniform accumulation crossover with an error certificate

Set $c_n = n^{-p}$ with $p > 1$. For $N \geq 2$ define the grouped basepoint tail

$$\mathcal{U}_N(t) = (-1)^{q+1} e^{-(q+1)t} t^{q+1} \sum_{n \geq N} n^{-p} \Psi_q(t/n). \quad (11.1)$$

This is a tail of the convergent grouped representation, not a tail obtained by separately summing the divergent canonical coefficients. Let

$$g_{p,q}(v) = v^{-p} \Psi_q(1/v), \quad I_{p,q}(c) = \int_c^\infty g_{p,q}(v) dv. \quad (11.2)$$

Both are positive for $v, c > 0$.

Theorem 11.1 (Uniform crossover and explicit two-sided error). *Let $t > 0$, $N \geq 2$, $c = N/t$, and*

$$\mathcal{E}_{N,t} = (-1)^{q+1} e^{(q+1)t} t^{p-q-2} \mathcal{U}_N(t).$$

Then

$$I_{p,q}(c) \leq \mathcal{E}_{N,t} \leq I_{p,q}(c) + \frac{g_{p,q}(c)}{t}. \quad (11.3)$$

The sharper centered estimate is

$$0 \leq \mathcal{E}_{N,t} - I_{p,q}(c) - \frac{g_{p,q}(c)}{2t} \leq \frac{-g'_{p,q}(c)}{8t^2}. \quad (11.4)$$

Consequently, if $N/t \rightarrow c_0 \in (0, \infty)$,

$$\mathcal{U}_N(t) \sim (-1)^{q+1} e^{-(q+1)t} t^{q+2-p} I_{p,q}(c_0).$$

The error estimates are uniform for c in compact subsets of $(0, \infty)$ and p in compact subsets of $(1, \infty)$, with fixed q .

Proof. A direct rescaling gives

$$\mathcal{E}_{N,t} = \frac{1}{t} \sum_{n \geq N} g_{p,q}(n/t). \quad (11.5)$$

Expanding (10.1),

$$g_{p,q}(v) = \frac{1}{q!} \sum_{k \geq 0} \frac{v^{-p-k}}{k!(q+k+1)}. \quad (11.6)$$

It is positive, decreasing, convex, and tends to zero. The same series shows that its derivative tends to zero and that its second derivative is integrable on every $[c, \infty)$. The ordinary decreasing-function rectangle bounds for (11.5) give (11.3).

For the sharper estimate, set $h = 1/t$. The trapezoidal error on $[a, a+h]$ is

$$\frac{h}{2}(g(a) + g(a+h)) - \int_a^{a+h} g(v) dv = \frac{1}{2} \int_a^{a+h} (v-a)(a+h-v)g''(v) dv.$$

Convexity makes it nonnegative, and the factor in front of g'' is at most $h^2/8$. Sum over all intervals beginning at c . Their total error is at most $\frac{h^2}{8} \int_c^\infty g''(v) dv = -h^2 g'(c)/8$. The left-endpoint sum in (11.5) is the infinite trapezoidal sum plus $hg(c)/2$. This proves (11.4). Uniformity follows from the displayed positive series and their derivatives on the stated compact parameter sets. $\square$

11.1 Evaluation formulas and the critical scaling

Termwise integration in (11.6) gives a useful convergent representation

$$I_{p,q}(c) = \frac{1}{q!} \sum_{k \geq 0} \frac{c^{1-p-k}}{k!(q+k+1)(p+k-1)}. \quad (11.7)$$

For the discrete quantity,

$$\mathcal{E}_{N,t} = \frac{t^{p-1}}{q!} \sum_{k \geq 0} \frac{t^k \zeta(p+k, N)}{k!(q+k+1)}, \quad (11.8)$$

where $\zeta(s, N) = \sum_{n \geq N} n^{-s}$ is the Hurwitz zeta tail for real $s > 1$. All terms are positive, and successive-term ratio bounds $t/(N(k+1))$ and $1/(c(k+1))$ give elementary geometric tail estimates for (11.8) and (11.7) once those ratios are below one.

The scale $N \asymp t$ has a precise interpretation. For a fixed exponent index, t/n becomes large, and the canonical exponential block resolves that term. For indices proportional to t , infinitely many nearly resonant blocks contribute together at scale $e^{-(q+1)t} t^{q+2-p}$. The moment threshold is $p = q + 2 = r$; at this threshold the extra power of t is zero. This does not remove the divergence of the canonical gauge, which at equality has a harmonic coefficient sum. The grouped tail, rather than the raw canonical sum, has the crossover limit.

11.2 Numerical illustrations

Table 1 uses $p = 5/2$ and $c = 1$. It includes dimensions in which the canonical gauge is divergent and nevertheless the stable representation and the proved bounds are valid.

Editorial note (ProveIt, 2026-09-29). For $r = 2$ ($q = 0$) and $c_n = n^{-p}$ the chain (9.1) is the accumulation model of the predecessor [2] (`eq:p-system`, with $S = \text{diag}(2, 0)$) after the input and the gauge entry are multiplied by x , which moves $\lambda = (1, 0)$ to S ; the predecessor's input also has the term $n = 1$, at the nonresonant exponent 1. Theorem 9.1 then gives its exact threshold $p > 2$ (`prop:threshold`), and Theorem 10.1 its renormalized solution h_p with $h_p(1) = 0$ (`thm:renorm`), up to the gauge of that one term, since $t\Psi_0(t/n) = n(e^{t/n} - 1)$ (the predecessor's Ψ_p is a different function). Moreover $g_{p,0}(v) = v^{1-p}(e^{1/v} - 1)$ is its f_p , so Theorem 11.1 at $q = 0$ is a second-order form, over $n \geq N$, of its fourth-order crossover `thm:crossover`; the two agree.

q	t	Scaled tail E	$I + g/(2t)$	Error bound
0	20	0.969362726	0.968260538	0.00165490768
0	80	0.936111703	0.936042753	0.00010343173
1	20	0.536929936	0.536260165	0.00100571307
1	80	0.517552067	0.517510165	6.2857067e-5
3	20	0.048558868	0.0484939007	9.75587143e-5
3	80	0.0467372268	0.0467331622	6.09741964e-6
5	20	0.00167737852	0.00167506571	3.47313388e-6
5	80	0.00161339799	0.00161325328	2.17070868e-7

Table 1: Crossover values and the upper bound in (11.4). The approximation is always below the scaled tail. Values are rounded; the inequalities are theorems, while these evaluations use ordinary high-precision arithmetic, not directed-rounding intervals.

12 Reproducibility, proof audit, and formalization targets

12.1 What was actually checked

The accompanying program `verification/verify.py` uses exact rational arithmetic for finite exponent supports and divided-power polynomials. Its recorded run, with seed 20260929, reports:

Check	Count
Scalar polynomial-inverse identities	105
Path-kernel degree and leading coefficient identities	210
Finite triangular systems	30
Differential-equation entry checks	540
Normalized-gauge entry checks	540
Constant-connection entry checks	540
Weighted path norm bounds	349
Explicit resonant-chain formulas	49
Crossover parameter cases	96

For each finite system, the code independently computes H by path expansion and by triangular recursion and checks equality. It verifies (4.1), constructs $(P, \mathcal{R})$, verifies (7.1), and checks that the connection between the two fundamental matrices is constant after the diagonal monomial shifts. The spectra include repeated eigenvalues, and the exponent samples include exact resonances. Finite tests also check the weighted inequality and the arbitrary-rank chain formula.

The numerical cases use $q \in \{0, 1, 3, 5\}$, $p \in \{1.5, 2, 2.5, 6.5\}$, $c \in \{1/2, 1, 2\}$, and $t \in \{20, 80\}$. They evaluate the positive series with analytic series-tail estimates and check both inequalities in Theorem 11.1. The executed environment was Python 3.13.5 and mpmath 1.3.0 at 75 decimal digits. Floating-point evaluations are not represented as interval certificates.

These tests do not prove the infinite-support theorem, the closed-graph argument, or the universal quantifiers. Those are conventional mathematical proofs in the preceding sections. No Lean theorem implementing this article’s new results was produced or kernel-checked.

12.2 Logical dependencies and delicate points

The core proof dependencies are

$$\begin{aligned} \text{finite path fibers} &\implies \text{formal path formula} \\ &\implies \text{exact weighted criterion} \\ &\implies \text{path gap equivalence.} \end{aligned}$$

The normalized log-free conclusion adds only the triangular normal-form construction and the constant-field comparison. The chain formulas and crossover are explicit calculations that can be checked independently of the functional-analytic necessity proof.

Several restrictions are indispensable to the stated argument. Acyclicity bounds the number of operators in a path and makes the set of paths finite. Independence of edge inputs permits isolation of one path. Positive real exponents make the bounded-total small-divisor argument transparent and fix the near-identity orientation. Well ordering, rather than local finiteness, provides finite exact coefficient fibers. The target of the closed-graph argument is a fixed-support Banach space. Exact resonances are handled by T_0 rather than by a fictitious division by zero.

The normalization comparison is an equivalence of membership, not an identity of operator norms. The all-radii result is stated for the log-free gauge and for the coefficient norm of its polynomial-logarithmic counterpart. The basepoint realization is a grouped analytic expression; its divergent pieces are never summed separately. Finally, the crossover tail is defined before taking a limit and is not conflated with a raw Hahn coefficient tail.

12.3 A realistic Lean decomposition

The existing support bridge [3] supplies the two-factor well-ordering and finite-decomposition interface. A formalization of the present results would add the following layers.

First, finite acyclic paths and their exponent tuple fibers can be handled without analysis. The polynomial operators T_β , their right-inverse identities, and the divided-power leading-coefficient lemma are finite-dimensional algebra. These are especially suitable initial targets.

Second, the formal path expansion and the triangular log-free normal form need explicit Hahn coefficient operations, a strong Euler derivation, and the polynomial-log constant-field lemma. Their proofs should expose the support bounds rather than rely on an informal field name.

Third, the weighted theorem requires fixed-support ℓ^1 spaces, coordinate continuity, closed graphs, and uniform boundedness. A formal statement should quantify over independent edge families. The exact operator-norm equality is a useful interface between the finite path algebra and the analytic layer.

Finally, the sharp converse needs the bounded-tuple small-divisor sequence and the leading-coefficient lower bound. The crossover proof can be formalized separately as a positive-series and trapezoidal-error theorem. This decomposition distinguishes existing repository support lemmas from new proof obligations; the former do not automatically verify the latter.

13 Further research questions

13.1 Cycles and infinite matrix-word languages

For a nontriangular perturbation, matrix words may have arbitrarily large length. Positive input exponents still bound the length of words below a fixed spectral cut, but they do not reduce the entire solution to finitely many multilinear maps. Can a finite critical-word condition, combined with a large-exponent majorant, characterize universal convergence? The first nontrivial case

is a two-vertex directed cycle with two accumulating supports. A proof must address repeated occurrences of the same input, not merely replace “path” by “walk” in Theorem 5.1.

13.2 Fixed Jordan blocks and zero-action edges

A fixed nilpotent part in A_0 behaves like a collection of zero-exponent edges. For a triangular constant matrix, finite path length survives, but some edge coefficients are fixed rather than independently variable. Determine the exact accessible small-divisor powers in terms of the Jordan filtration and the variable-edge pattern. The chain in Section 9 suggests that logarithmic height, not merely the number of near-resonances, is the relevant weight exponent.

13.3 Coupled matrix directions and relations among words

The four-dimensional cancellation example shows that a graph alone is insufficient when coefficients multiply prescribed linear combinations of matrix units. A natural replacement is a word algebra modulo the matrix relations, with a kernel attached to each surviving coefficient tensor. Find an exact independent-parameter criterion that remains valid after these relations are imposed. Even two fixed nilpotent generators with a nontrivial quadratic relation would be a useful first classification.

13.4 Optimal factorized weights

The exact criterion asks for $|Q_p(\alpha)| \leq C_p \prod w_e(\alpha_e)$. Which minimal or Pareto-optimal families of edge weights satisfy all these inequalities? For finitely sampled supports, taking logarithms leads to finite inequality systems; for accumulating supports, one must control asymptotic factorization of singular path kernels. Such a result would turn the criterion into a systematic regularity-design tool rather than a post hoc convergence test.

13.5 Moving spectra through an accumulation point

Let a spectral difference move by a parameter δ through a left accumulation point. Canonical normalizers have denominators such as $\delta - 1/n$, while basepoint integrals can remain regular in δ . Determine a uniform theorem when t/n and δt compete. Exact resonance, near resonance, and the accumulation boundary should be treated in a single analytic representation, not by separately divergent formulas.

13.6 Multiplicative realization and iterated integrals

The basepoint construction is compatible with ordinary analytic matrix products because it solves a genuine differential system. This does not yet define a product-preserving summation map on the underlying formal Hahn algebra. Can one identify a restricted algebra of accumulating-support symbols whose realizations are closed under products and satisfy the shuffle identities of iterated integrals? The first target should include two interacting resonant chains, where nonzero products force compatibility conditions beyond a scalar subtraction rule.

13.7 Triangular nonlinear systems and tree kernels

For polynomial triangular differential systems, substitutions branch into rooted trees rather than simple paths. Does a finite-tree analogue of the weighted kernel criterion hold, and when do diagonal coefficient identifications obstruct its necessity? Polarization may recover some independence, but repeated use of one input and coinciding exponents must be handled

explicitly. This would connect the present classification to nonlinear transseries reversion without presupposing convergence.

13.8 Effective presentation of infinite supports

The criterion is semantic: it refers to left accumulation of exact sumsets. For which computable descriptions of E_e can the condition be decided or certified? Finite rational supports are elementary, but arbitrary computable increasing sequences can encode difficult limiting questions. A useful next result would distinguish finite symbolic certificates, semidecidable obstruction searches, and genuinely decidable presentation classes.

13.9 Irregular exponential sectors

Our systems are regular singular in the Euler sense and the expansions use real powers and logarithms. Irregular systems introduce exponential sectors and potentially factorial coefficient growth. Can one separate an accessible finite-spectral accumulation obstruction from a Borel-plane summability obstruction in one explicit triangular irregular model? A successful theorem should retain two distinct norms, rather than call both mechanisms “resurgence.”

13.10 Kernel-checked necessity, not only finite examples

The most valuable formal target is the full implication from universal convergence to bounded path kernels. Its proof exposes the quantifier that finite numerical experiments cannot establish. A staged implementation should first verify the scalar and finite-path algebra, then the fixed-support closed-graph lemma, and finally the exact gap equivalence. The supplied rational verifier is a reproducibility aid, not a substitute for this goal.

14 Conclusion

For acyclic matrix patterns, universal absolute convergence is governed by what the differential system can actually multiply. The complete exponent semigroup is too coarse, and inspecting each edge separately is too weak. The correct invariant is the family of endpoint-labelled path sumsets. The exact weighted refinement retains the full polynomial-logarithmic kernel and therefore detects powers of small denominators caused by resonant descendants.

This resolves a concrete fixed-support question from the repository for a substantial precisely specified class, gives explicit counterexamples to two natural simplifications, and quantifies an arbitrary-rank resonance loss. The stable chain realization also demonstrates that divergence of a canonical normalizer is not nonexistence of analytic solutions or loss of every finite asymptotic prefix. The remaining questions concern the extra algebra introduced by cycles, coupled directions, nonlinear trees, and irregular exponential scales. Those are extensions of the proved boundary, not assumptions silently included in it.

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